

Part H: Final Analysis & Reporting

PART H • Q25 FINAL REPORT (ANALYSIS & INTERPRETATION)



Q25. Final report covering the best-performing model and justification, impact of regularization, role of cross-validation in model stability, comparison of linear vs non-linear regressors, and business interpretation of results. All figures come from the Robust Regression Engine notebook (Parts B–G).

1. Best-Performing Model and Justification

Lasso Regression ($\alpha=100$) is the best-performing model, with **Ridge ($\alpha=1$)** practically tied and **Random Forest** a very close second.

Model	Val R^2	RMSE (₹)	MAE (₹ lakh)	Fit
Lasso	0.9187	25.58 lakh	19.59	Good Fit
Ridge	0.9187	25.58 lakh	19.59	Good Fit
Random Forest	0.9143	26.28 lakh	19.37	Good Fit
Decision Tree (tuned)	0.8835	30.64 lakh	23.25	Good Fit
SVR (RBF, tuned)	≈ 0.00	89.86 lakh	69.93	Underfitting

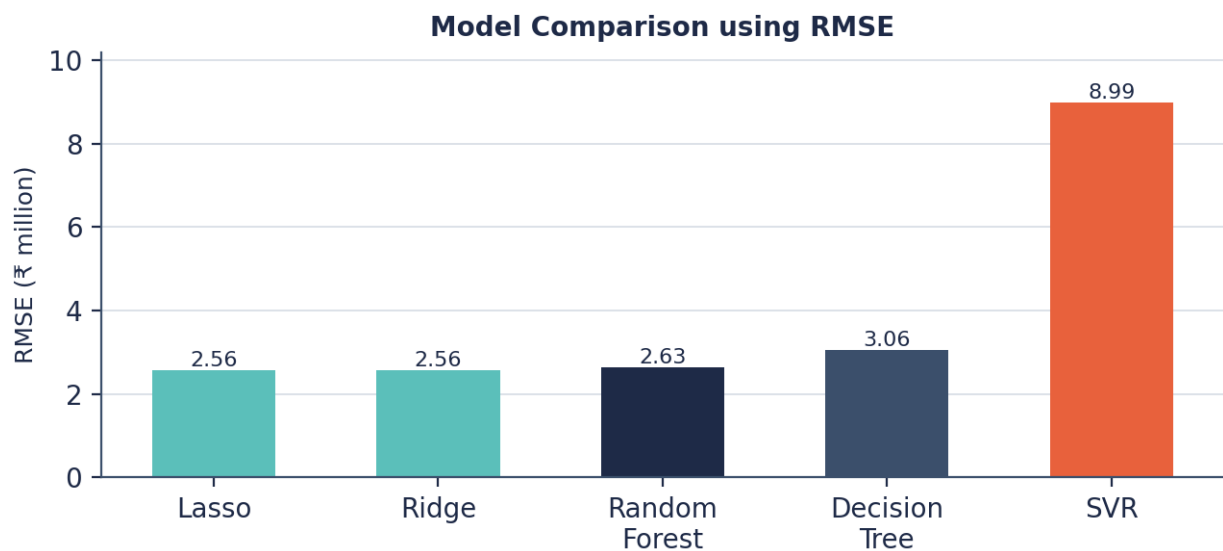


Figure 1: Test RMSE of all models. Lower is better.

Justification

- Lowest RMSE ($\approx ₹25.58$ lakh) and highest validation R^2 (0.9187) among all models.
- **No overfitting:** Train $R^2 \approx 0.916$ vs Validation $R^2 \approx 0.919$.
- **Simple and interpretable:** each coefficient shows how a feature moves the price, and Random Forest offers only a marginal gain (lowest MAE) at higher complexity.

2. Impact of Regularization

Model	Setting	Train MSE ($\times 10^{12}$)	Val MSE ($\times 10^{12}$)
Ridge	$\alpha = 1$ (tuned)	6.253	6.545
Lasso	$\alpha = 1$	6.253	6.544
Lasso	$\alpha = 100$ (tuned)	6.253	6.544
Decision Tree	Unrestricted	0.000	11.419
Decision Tree	Depth 5, min leaf 5	7.503	9.386

Insight

- **Linear models:** tuning α from 1 to 100 barely changed the errors. With only 9 features and a strong signal, the data needs little regularization, and Lasso did not zero out any feature.
- **Where it mattered:** the unrestricted tree had Training MSE = 0 ($R^2 = 1.0$) but Validation R^2 of only 0.858, a clear case of overfitting. Constraining its complexity cut validation MSE by $\sim 18\%$.
- Regularization is therefore a safeguard here: it keeps models stable and generalizable rather than delivering a large accuracy jump.

3. Role of Cross-Validation in Model Stability

Strategy	Average MSE ($\times 10^{12}$)	Note
K-Fold (5)	6.316	Folds range 5.83 to 7.13
Stratified K-Fold	6.300	Target binned into 5 quantiles
Leave-One-Out	6.298	Same estimate, highest cost
Time Series Split	6.379	Most realistic (past $\rightarrow$ future)

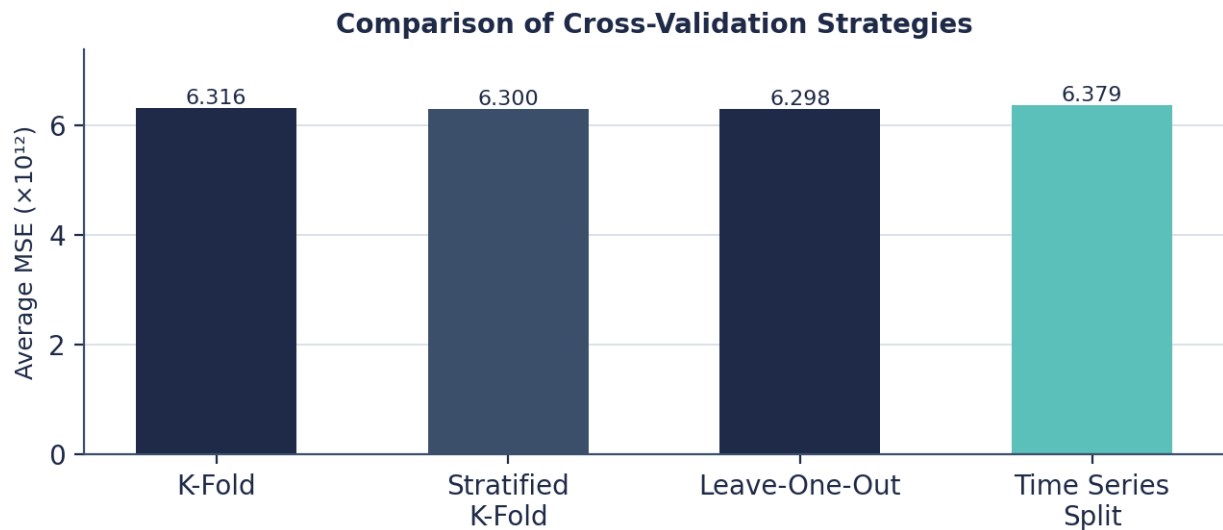


Figure 2: Average MSE stays within $\sim 1.3\%$ across all four strategies.

Insight

- All four strategies agree (6.30 to 6.38×10^{12}), so the performance estimate is **stable and reliable**.
- Individual K-Fold scores vary noticeably from fold to fold, so relying on a single train-test split would be risky.
- Cross-validation (GridSearchCV, $cv=5$) was also used to tune α for Ridge and Lasso and the parameters of SVR.
- Time Series Split shows no major drift over time, and 5-fold K-Fold is sufficient in practice.

4. Comparison of Linear vs Non-Linear Regressors

Type	Model	Val R^2	RMSE (₹ lakh)	Behaviour
Linear	Ridge / Lasso	0.9187	25.58	Stable, interpretable
Non-linear	Random Forest	0.9143	26.28	Lowest MAE, some larger errors
Non-linear	Decision Tree	0.8835	30.64	Step-wise predictions
Non-linear	SVR (RBF)	≈ 0.00	89.86	Failed: unscaled target

Insight

- Simple linear models beat the non-linear ones, which suggests the price-feature relationship in this dataset is **mostly linear**. Complex models add little here.
- Random Forest comes close because averaging many trees reduces variance (~26% lower MSE than a single tree).
- SVR failed because the target (₹10⁷ scale) was not scaled, not because of the kernel. Scaling the target or using a much larger C is the fix.

5. Business Interpretation of Results

Prediction accuracy. The best model is off by about **₹19.6 lakh on average** (MAE), roughly 9–10% of the mean price (₹2.07 crore). This is reasonable accuracy for price estimation and screening, but not a replacement for a property valuation.

Feature	Effect on price	Approx. effect, others held constant
area_sqft	Increases price	+₹11,900 per sq ft
location_score	Increases price	+₹20.8 lakh per score point
bathrooms	Increases price	+₹2.5 lakh per bathroom
bedrooms	Increases price	+₹2.2 lakh per bedroom
property_age	Lowers price	−₹52,800 per year
crime_rate_index	Lowers price	−₹66,200 per index point
distance_city_km	Lowers price	−₹4,600 per km

Approximate effects are the Ridge coefficients (per 1 standard deviation) divided by each feature's standard deviation from the dataset summary.

Key takeaways for decision-making

- **Size and location drive value most.** area_sqft and location_score have the largest positive impact, so they should anchor any pricing or investment decision.
- **Older properties, higher crime and greater distance from the city reduce price**, so these are the main discount factors when valuing a property.
- **near_school and near_metro have the smallest impact**, so they should not be over-weighted in pricing.
- **Recommended model:** use Lasso/Ridge for estimates because they are accurate, stable and explainable to non-technical stakeholders, with Random Forest as a cross-check. SVR should not be used until the target is scaled.

Overall

The relationship between price and features in this dataset is mostly linear, so **Lasso/Ridge are the most balanced and reliable choice**, with Random Forest as a strong alternative.